$$P(-z_{\alpha/2} < z < z_{\alpha/2}) = 1-\alpha$$

$$P\left(-z_{\alpha/2} < \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} < z_{\alpha/2}\right) = 1-\alpha$$

$$\downarrow$$

$$\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$100(1-\alpha)\%$ $p_1 - p_2$: $\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$



$$P(-t_{\alpha/2, n-1} < T_{n-1} < t_{\alpha/2, n-1}) = 1-\alpha$$

$$P\left(-t_{\alpha/2, n-1} < \frac{\bar{X} - \mu}{S/\sqrt{n}} < t_{\alpha/2, n-1}\right) = 1-\alpha$$

$$P\left(-t_{\alpha/2, n-1}\frac{S}{\sqrt{n}} \overset{-\bar{X}}{<} \bar{X} - \mu \overset{-\bar{X}}{<} t_{\alpha/2, n-1}\frac{S}{\sqrt{n}}\right) = 1-\alpha$$

$$\downarrow -1 \cdot *$$

$$P\left(\bar{X} - t_{\alpha/2,n-1} \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2,n-1} \frac{S}{\sqrt{n}} \right) = 1-\alpha$$